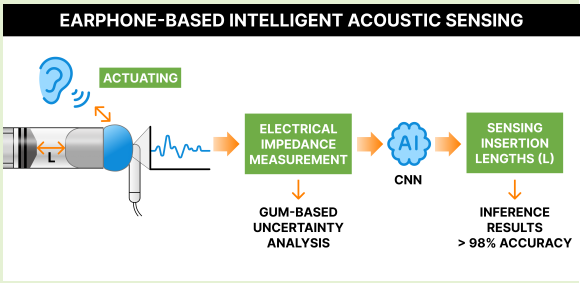
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Impedance-Based Acoustic Load Inference Using Intelligent Speaker Sensors

Noori Kim*, *Senior Member, IEEE*, Catherine H. S. Jeon, and Mingsian R. Bai, *Senior Member, IEEE*

***Abstract*— This work demonstrates the feasibility of repurposing speakers (earphones), originally designed as actuators, as intelligent sensors to sense acoustic loads such as insertion lengths attached to the speakers by exploiting speaker-coil electrical impedance as an information-bearing signal for acoustic load inference. By leveraging electroacoustic coupling governed by motional impedance theory, variations in electrical impedance are measured under different acoustic loading conditions and used as input features for artificial intelligence (AI) modeling, revealing a strong relationship between acoustic conditions and electrical properties. The resulting impedance spectra are fed into a convolutional neural network (CNN), achieving inference accuracies exceeding 98% across multiple evaluated acoustic conditions. Measurement robustness is supported by an uncertainty analysis that quantifies key sources of frequency-dependent variability, ensuring reproducibility of the sensing pipeline. By integrating electroacoustic principles with data-driven inference, this work provides a robust theoretical and system-level framework for earphone-based intelligent sensing, enabling compact and energy-efficient diagnostic capabilities at the edge.**



***Index Terms*— Electroacoustics, Speaker-as-Sensor, Impedance-Based Sensing, Deep Learning, Intelligent Sensors**

I. Introduction

THE rapid expansion of intelligent sensing systems has driven exploration beyond traditional sensor–signal–processor pipelines toward unconventional sensing paradigms. This paradigm shift enables devices originally designed as actuators or interfaces to be repurposed as sensing elements, in which raw physical responses are no longer treated merely as intermediate measurements but are instead regarded as information-bearing signals that can be transformed into actionable information through data-driven inference at the edge. Such approaches open new and broaden opportunities for compact, low-cost, and energy-efficient sensor architectures.

Electrical impedance sensing provides a particularly attractive foundation for such architectures, as impedance variations inherently reflect physical interactions across electrical, mechanical, and acoustic domains. Changes in material properties [1], [2], fluid composition [3], and structural conditions [4] have all been shown to manifest as measurable impedance signatures. However, due to the nonlinear and coupled nature of these interactions, analytical modeling alone often becomes intractable. In this context, artificial intelligence (AI) offers a powerful mechanism to directly map impedance responses to semantic information, bypassing the need for fully parameterized physical models.

Recent advances in AI-enabled sensing have demonstrated that data-driven models can effectively learn complex relationships between impedance signatures and physical states, enabling automated detection, characterization, and inference in diverse application domains [5]. For example, impedance-based learning frameworks have been applied to structural monitoring and machinery diagnostics, where convolutional neural networks (CNNs) and other deep learning architectures extract informative features directly from raw electrical responses [6], [7]. These studies highlight the growing role of system–algorithm co-design, in which sensing hardware and learning models are jointly optimized to achieve robust inference under real-world conditions.

Despite these advances, impedance-based intelligent sensing has seen limited exploration in electroacoustic systems. Nevertheless, strong conceptual parallels exist with electromechanical impedance (EMI) techniques widely used in structural health monitoring, where changes in mechanical or acoustic loading are reflected in the electrical impedance of a transducer [8]. Both EMI and electroacoustic sensing rely on motional impedance coupling, making impedance spectroscopy a natural interface between physical interaction and electrical response [9]. These principles suggest that electroacoustic transducers may serve as viable sensing elements, even when not originally designed for measurement purposes.

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In hearing and in-ear acoustic applications, accurate characterization of acoustic loading is strongly influenced by ear-canal geometry and insertion depth. Variations in probe or earphone insertion depth alter sound propagation and acoustic impedance, directly affecting measurement reliability and user comfort [10], [11]. Existing approaches for controlling insertion conditions rely on visual inspection, reinsertion, or microphone-based sealing checks [12]–[14], which are often subjective and sensitive to individual anatomical differences. These limitations motivate the development of sensing approaches that can directly infer acoustic loading without requiring additional sensors or manual intervention.

Building on the concept of repurposing actuators as sensors, prior work first demonstrated the feasibility of using earphones as impedance-based sensing elements for acoustic load classification [15], [16]. While promising, these early studies were constrained by image-based impedance representations, large neural network architectures, and limited datasets. The present work advances this direction by directly leveraging raw speaker-coil impedance as an information-bearing signal, enabling efficient inference through a customized CNN architecture.

In this study, a speaker is explicitly engineered to function as both an actuator and a sensor within an integrated electroacoustic sensing pipeline. Although a syringe-based cavity (acoustic loads used in this study) is used to emulate ear-canal insertion depth, this simplified geometry retains analytical relevance, as the ear canal is commonly modeled as a cylindrical cavity at low frequencies [17]. More broadly, the proposed approach generalizes to other applications requiring low-cost, non-invasive estimation of insertion depth or cavity length in cylindrical structures. To interpret measurement uncertainty, this work also leverages electro–mechano–acoustic formulations that relate acoustic loading to electrical impedance. These are used in the Discussion to explain how variations in acoustic conditions propagate and influence the observed uncertainty. By integrating electroacoustic principles with AI-driven inference, this work introduces a new class of speaker-based intelligent sensors. The proposed framework demonstrates how commonly available electroacoustic transducers can perform edge-level acoustic load inference without dedicated sensing hardware, contributing to the broader vision of intelligent, adaptive, and resource-efficient sensor systems.

II. Methods

A. Experimental Set-up

The experiment was designed to measure the electrical impedance of earbud speakers under controlled acoustic loading conditions, using a syringe-based enclosure as a simplified and repeatable proxy for small cavity structures such as the ear canal. The syringe-based setup enabled well-controlled variation of key acoustic parameters, including enclosure length and speaker positioning, ensuring consistent and repeatable measurement conditions.

Four miniature loudspeakers were characterized based on their DC impedance and the mechanical gap between the diaphragm and the ear-tip reference plane. The DC impedance was measured with a Digital Multimeter (Keysight EDU34450A).

- Speaker A (Airline A): 32.59 Ω with a 3 mm gap
- Speaker B (Airline B): 31.18 Ω and 2 mm gap
- Speaker C (Sennheiser CX80S): 27.35 Ω and 2 mm gap
- Speaker D (Amazon Singapore): 33.18 Ω and 2 mm gap.

To ensure a consistent fit within the syringe enclosure, the ear-tips were attached to the main body of the speakers (Figure 1). These ear-tips introduced a measurable gap between their edge, which ranged from 2 to 3 mm, and the speaker grille (depending on each speaker), which was accounted for during measurements. The gaps were measured with a caliper. Also, the DC impedance from 27 Ω to 33 Ω; consistent with typical low-power earphone drivers.

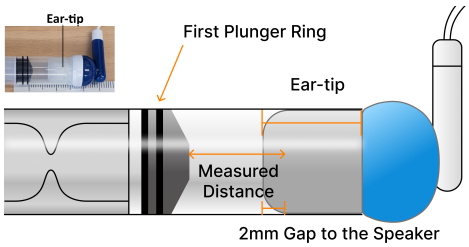


Fig. 1: Example of the enclosure setup with the 2 mm gap to the speaker grille, which is included as part of the measured distance. A picture of the speaker used in this study, indicating the ear-tip part, is shown as a physical reference.

After the speakers were fitted into the syringe, the syringe length was adjusted to simulate varying enclosure dimensions. A marker line on the syringe body served as a reference for ear-tip placement, while a ruler was used to ensure coarse positioning, as illustrated in Figure 1. The internal cavity length defining each enclosure condition was measured using a stainless-steel vernier caliper (accuracy ± 0.025 mm). Because the syringe inner diameter was smaller than the caliper jaw opening, the measurement was performed externally while visually referencing the internal geometry through the transparent syringe wall.

For each target length, the plunger position was first adjusted using the ruler and then refined by measuring the axial distance between the syringe tip and the inner face of the plunger, as observed through the transparent syringe body. The caliper jaws were aligned parallel to the syringe axis and placed on external surfaces corresponding to the internal reference points, enabling the internal cavity length to be read directly without inferring values from external wall thickness. Measurements were repeated to confirm consistency within the stated caliper accuracy. Each configuration was visually inspected to verify a consistent seal between the ear tip and syringe wall. For example, to achieve a 9 mm enclosure length for speaker D, the syringe plunger was set 7 mm from the marker line to account for the 2 mm gap.

To prevent measurement artifacts, the speakers were removed after each adjustment of the syringe plunger. Following each repositioning, the system was allowed to stabilize for several seconds to equalize internal pressure before the

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next impedance sweep was initiated. This stabilization period reduced pressure-induced perturbations that could otherwise alter the effective acoustic load. Each sweep was performed three times consecutively to confirm measurement repeatability, and any anomalous data showing discontinuities or unexpected phase jumps were discarded and remeasured.

The measurement circuit featured two input options: a terminal block for bare wire connections and a 3.5 mm standard audio Tip-Ring-Sleeve (TRS) input for stereo earbuds. A switch allowed selection between the left and right channels for the TRS connection. This flexibility ensured compatibility with various speaker configurations. Figure 2 shows the schematic of the measurement circuit with designated voltage nodes and the method used to compute the input impedance.

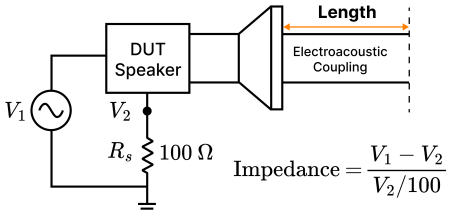


Fig. 2: Measurement circuit schematic. V_1 applies a 20 Hz–20 kHz, 500 mV-peak sinusoidal sweep.

B. Data Collection

Impedance measurements were acquired using the Digilent Analog Discovery 2 (AD2) device [18], controlled via the WaveForms software. The excitation source analog outputs are driven by rail-to-rail AD8067 amplifiers. The output impedance is effectively 0Ω under normal operation, limited only by a protective Positive Temperature Coefficient thermistor that engages under fault conditions. Thus, the voltage drop at the source is negligible for the 500 mV RMS excitation, allowing the source to be treated as an ideal voltage generator in the impedance calculations.

The measurement setup employed a stepped-sine excitation protocol, consisting of 501 discrete, linearly spaced single-tone frequencies spanning 20 Hz to 20 kHz. In contrast to broadband excitation, the stepped-sine approach concentrates energy at each test frequency, resulting in a high effective Signal-to-Noise Ratio (SNR) in the steady-state response without requiring exponential sweep processing. Each frequency was applied sequentially, and the corresponding steady-state response was recorded. Signal generation and acquisition were performed using an AD2, with both the waveform generator and oscilloscope operating at a sampling rate of 100 MSs. Each excitation signal had a source voltage amplitude of 500 mV (peak) with zero DC offset to ensure within the small-signal linearity regime (≤ 1 V RMS excitation).

It is also to minimize nonlinear distortion and ensure that measured impedance variations reflect acoustic loading rather than transducer nonlinearity. In place of a 200 mΩ load in parallel, a 100 Ω resistor (R_s)is connected in series with the device under test (DUT), serving as a reference for impedance calculations. This series configuration facilitates analysis of the system's impedance and phase behavior, providing greater precision in quantifying both magnitude and phase. The measurement process involved hardware setup, executing a frequency sweep, and exporting as a CSV file.

To streamline data collection, a custom data collection program was developed using Tkinter in Python [19]. This interfaces with the AD2 device, automating file naming, repeated measurements, and dataset organization. The program allows the selection of the current speaker, the distance being measured, the number of repetitions, as well as the other settings mentioned previously.

Data was collected for four distinct speakers across 14 enclosure lengths. Measurements were conducted at intervals of 3 mm, starting at 5 mm up to 29 mm. These were also supplemented by additional key distances from previous studies [16], notably 9 mm, 24 mm, and 39 mm, as well as Open and Blocked conditions.

For each configuration, 100 runs were performed, where each run represents one complete impedance measurement cycle consisting of sequential stepped-sine excitation across the full frequency range (501 discrete single-tone frequencies ranging from 20 Hz to 20 kHz). It results in a total dataset of 4 speakers * 14 distances * 100 runs = 5600 total measurement samples for use with machine learning (ML). Each sample comprises 501 data rows with the following fields as columns:

- Frequency (Hz): Frequency point for the sweep.
- Phase (in degree θ): Impedance angle at the frequency.
- $|Z|$ (Ω): Impedance magnitude.
- R (Ω): Real part of the impedance.
- X (Ω): Reactive component of the impedance.

To add a computational description of each collected parameter, let $V_{\mathrm{DUT}}(f)$ and $I_{\mathrm{DUT}}(f)$ denote the complex voltage and current spectra at frequency f, obtained from the swept-sine measurement. The complex input impedance of DUT, also shown in Fig. 2 is computed as:

$$Z(f) = \frac{V_{\mathrm{DUT}}(f)}{I_{\mathrm{DUT}}(f)} = |Z(f)|e^{j\angle Z(f)} = \frac{V_1 - V_2}{V_2/R_s} = R_s\frac{V_1 - V_2}{V_2}, \tag{1}$$

where $R_s = 100\Omega$ in our setting. The magnitude and phase are related to the resistive and reactive components as $R(f) = |Z(f)|\cos\big(\angle Z(f)\big)$ and $X(f) = |Z(f)|\sin\big(\angle Z(f)\big)$ where $\Re\{\cdot\}$ and $\Im\{\cdot\}$ denote the real and imaginary parts, respectively, and $\arg(\cdot)$ denotes the phase angle.

C. Data Preprocessing

The electrical impedance data for each speaker were averaged across 100 runs and plotted in Fig. 3, showing magnitude responses across frequency. Each trace was normalized to the 20 Hz value to enable relative comparison across conditions, reduce systematic effects, and emphasize frequency-dependent behavior. At 20 Hz, the impedance varied by 3–5% across 100 measurements for each condition.

For each speaker system, the four fields—Phase, $|Z|$, R, and X—were extracted and concatenated along the channel axis to form a 4-channel feature representation. These variables are frequency-dependent components in the impedance response, defined over an identical frequency grid across all samples.

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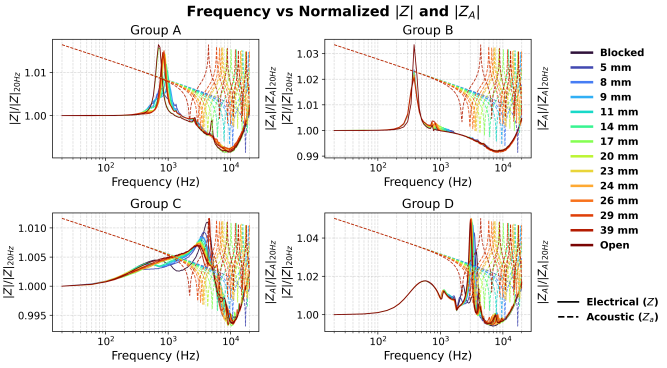


Fig. 3: Frequency vs. Magnitude $|Z|$ and $|Z_A|$ of Speaker Data

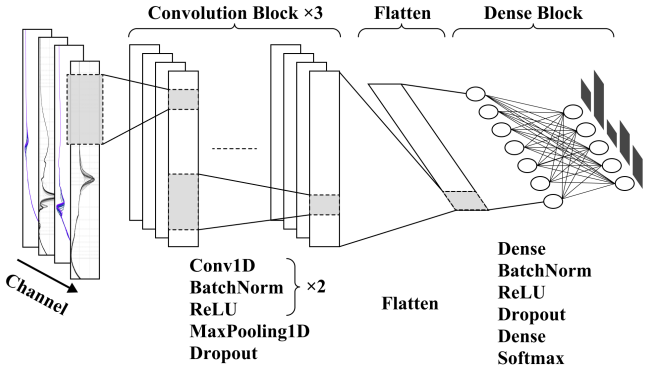


Fig. 4: Overview of CNN Model Architecture

Although $|Z|$ and Phase $vs.$ R and X representations are mathematically equivalent, both were included to provide complementary feature representations, which empirically improved model performance.

We denote this input as a 3D tensor $T \in \mathbb{R}^{N\times L\times C}$ with N training samples, sequence length $L = 501$, and channels $C = 4$. For sample $i \in \{1,\dots,N\}$, position $\ell \in \{1,\dots,L\}$ and channel $c \in \{1,\dots,C\}$, let $T_{i,\ell}^{(c)}$ be the value at (ℓ, c) in sample i. We compute the mean $\bar{T}_\ell^{(c)}$ and standard deviation $SD_\ell^{(c)}$along the sample axis, independently for each location (ℓ, c), using only the training set, as follows:

$$\hat{T}_{i,\ell}^{(c)} = \frac{T_{i,\ell}^{(c)} - \bar{T}_\ell^{(c)}}{SD_\ell^{(c)}}, \quad \bar{T}_\ell^{(c)} = \frac{1}{N}\sum_{i=1}^{N} T_{i,\ell}^{(c)}, \tag{2}$$

$$SD_\ell^{(c)} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(T_{i,\ell}^{(c)} - \bar{T}_\ell^{(c)}\right)^2}. \tag{3}$$

As the composition of speakers in the training data differs across experimental settings, $\bar{T}_\ell^{(c)}$ and $SD_\ell^{(c)}$ reflects either cross-speaker variability when the dataset contains multiple speakers, or within-speaker variability when the model is trained on a single speaker. However, in all settings, this zero-centered normalization ($\hat{T}_{i,\ell}^{(c)}$) significantly helps stabilize the training of the deep learning model.

For the target labels $Y \in \mathbb{R}^N$, we only converted categorical values into one-hot encoding vectors. Finally, the dataset was then randomly shuffled and split into 80% training, 10% validation, and 10% testing subsets, following the traditional deep learning training procedure.

D. Model Training

The task is formulated as a supervised classification problem with target labels corresponding to acoustic path lengths of $5, 8, 9, 11, 14, 17, 20, 23, 24, 26, 29,$ and 39 mm, along with two additional classes, Blocked and Open. Three scenarios, namely "Speaker and Length," "Only Length," and "Individual Speakers," were evaluated to assess the model's ability to distinguish among the four speakers (A–D) for each length (Section III).

For the model architecture, we employed a CNN-based classification model. The Keras Sequential framework is used with 1) six convolution layers and 2) two fully connected layers. Specifically:

- **Convolution Block**: Three Convolution blocks were implemented for rich feature representation from the input data. Each block applies the following operations to the l-th block input x_l, producing the output x_{l+1}.

$$\begin{aligned} h_1 &= \mathrm{ReLU}(\mathrm{BatchNorm}(\mathrm{Conv1D}(x_l))),\\ h_2 &= \mathrm{ReLU}(\mathrm{BatchNorm}(\mathrm{Conv1D}(h_1))),\\ h_3 &= \mathrm{MaxPool1D}(h_2),\\ x_{l+1} &= \mathrm{Dropout}(h_3). \end{aligned}$$

  Each block gradually increases the number of convolution filters; specifically, 16, 32, and 64 in the first, second, and last blocks, respectively.
- **Dense Block**: The dense block serves as the multi-class classifier, structured as follows:

$$\begin{aligned} h_1 &= \mathrm{Dropout}(\mathrm{ReLU}(\mathrm{BatchNorm}(\mathrm{Dense}(x_l)))),\\ x_{l+1} &= \mathrm{Softmax}(\mathrm{Dense}(h_1)). \end{aligned}$$

The model was trained using categorical cross-entropy as the loss function, using a batch size of 128 and 20 training epochs. Optimization was performed with the Adamax optimizer and the label smoothing of 0.2. The overall model architecture is depicted in Figure 4.

E. Acoustic Impedance Modeling of Cylindrical Tubes

To distinguish from the measured electrical impedance, the acoustic impedance of each syringe cavity is modeled as a lossy cylindrical tube [9]. This model is introduced to support the interpretation of the measured impedance responses and associated uncertainty by relating variations in cavity geometry to changes in acoustic loading. The acoustic impedance of the tube is expressed as

$$Z_{\mathrm{A}} = Z_0 \coth(\gamma L), \tag{4}$$

where Z_0 is the characteristic impedance, γ is the complex propagation constant accounting for viscous and thermal losses, and L is the tube length, following the formulation of Keefe [20]. Excluding the boundary conditions (blocked and open), the simulated $|Z_A|$ from the remaining 12 cavity lengths are plotted in Fig. 5 normalized at 20 Hz. Also, for easy characteristics comparison between measured electrical impedance of the speaker with various tube lengths, and this $|Z_A|$, they are plotted on the same plane in Fig. 3.

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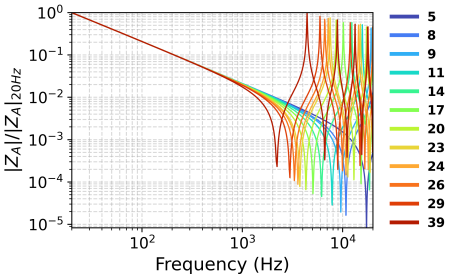


Fig. 5: Acoustic impedance magnitude of cylindrical tubes for 12 cavity lengths (excluding blocked and open conditions).

TABLE I: Condition-marginal Type A uncertainty components for $|Z|$ across Blocked, 14 mm, and Open conditions (repeatability + unit + length).

Freq. (Hz)	u_{repeat} (RMS)	SD_{unit}	SD_{length}	u_c	U_{95}
400	0.0041	3.6	0.058	3.6	11
800	0.0041	3.5	0.13	3.5	11
1600	0.0041	3.4	0.11	3.4	11
3200	0.0041	3.6	0.35	3.6	11
6400	0.0039	3.4	0.11	3.4	11

III. Results

A. Evaluation Metrics

We evaluated the proposed deep learning model using traditional ML classification metrics as follows:

$$\text{Accuracy} = \frac{\text{True Positive + True Negative}}{\text{Total cases}},$$
$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive + False Positive}},$$
$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive + False Negative}},$$
$$\text{F1-Score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}.$$

Here, the True Positives and True Negatives denote correct classifications, while the False Positives and False Negatives indicate misclassified negative and positive instances, respectively. Since the F1-score represents a comprehensive measure that integrates both precision and recall, we report accuracy and F1-score as the evaluation metrics. For reliability, all experiments were repeated three times, and the results are reported as the average values.

B. Uncertainty Analysis

A comprehensive uncertainty evaluation was conducted following the *Guide to the Expression of Uncertainty in Measurement (GUM)* [21]. Both statistical (Type A) and systematic (Type B) sources were considered to ensure metrological traceability and reproducibility of the impedance measurements. This is to support the reliability and interpretability of the proposed sensing framework to quantify measurement variability and its impact on model performance.

To visualize statistical dispersion and frequency-dependent variability in the original dataset, Figure S3 in the online Supplementary Material shows the distributions of the measured impedance features ($|Z|$, phase, R, and X) across all speakers and syringe lengths, including the mean and standard deviation (SD) from 100 repeated measurements.

TABLE II: Condition-marginal Type A uncertainty components for Phase across Blocked, 14 mm, and Open conditions (repeatability + unit + length).

Freq. (Hz)	u_{repeat} (RMS)	SD_{unit}	SD_{length}	u_c	U_{95}
400	0.00090	0.51	0.12	0.53	1.6
800	0.0010	0.11	0.11	0.16	0.40
1600	0.0013	0.075	0.022	0.078	0.23
3200	0.0017	0.26	0.56	0.62	2.0
6400	0.0021	0.71	0.058	0.71	2.3

TABLE III: Condition-marginal Type A uncertainty components for R across Blocked, 14 mm, and Open conditions (repeatability + unit + length).

Freq. (Hz)	u_{repeat} (RMS)	SD_{unit}	SD_{length}	u_c	U_{95}
400	0.0041	3.6	0.058	3.6	11
800	0.0041	3.5	0.13	3.5	11
1600	0.0041	3.4	0.11	3.4	11
3200	0.0041	3.6	0.36	3.6	11
6400	0.0039	3.4	0.11	3.4	11

1) Condition-marginal Type A uncertainty (system + unit + length): Type A uncertainties were then derived from repeated measurements (repeat) across multiple speakers (unit) and load conditions (length) to quantify repeatability, inter-unit variability, and load-dependent variation.

For the Type A analysis, we report three acoustic-load conditions that bracket practical operation: Blocked (high-impedance limit), 14 mm (representative average ear-canal length), and Open (low-impedance vented boundary). Also, anchor frequencies $f \in 400, 800, 1600, 3200, 6400\text{Hz}$ are selected because they cover speech- and environment-critical bands: lower/mid formant regions (400–1600 Hz) and the upper-mid/high bands (3.2–6.4 kHz) that carry intelligibility cues, frication energy, and typical transducer features. This choice yields concise, application-relevant metrological summaries while remaining tied to the broader full-band results. Including $|Z|$, Type A evaluations for the phase, R, and X components were performed to ensure metrological completeness; detailed results for these supplementary parameters are provided in the online Supplementary Materials with 12 separate tables (Table S1–S12).

For each speaker (A–D) and anchor frequency,

- Mean $|\bar{Z}|(f) = \frac{1}{N}\sum_{i=1}^{N}|Z|_i(f)$, the average of N repeated runs under the stated load.
- Repeatability SD

$$SD_{\text{repeat}}(f) = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}\left(|Z|_i(f) - |\bar{Z}|(f)\right)^2},$$

- Type A uncertainty associated with the reported mean

$$u_A(f) = \frac{SD_{\text{repeat}}(f)}{\sqrt{N}}.$$

$N = 100$ runs were collected per condition, and the per-condition Type A tables report $|\bar{Z}|(f)$ together with $u_A(f)$ (standard deviation of the mean). These per-condition Type A

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TABLE IV: Condition-marginal Type A uncertainty components for X across Blocked, 14 mm, and Open conditions (repeatability + unit + length).

Freq. (Hz)	u_{repeat} (RMS)	SD_{unit}	SD_{length}	u_c	U_{95}
400	0.00050	0.29	0.068	0.30	0.90
800	0.00060	0.059	0.061	0.085	0.22
1600	0.00070	0.046	0.012	0.048	0.14
3200	0.00090	0.15	0.33	0.36	1.2
6400	0.0011	0.44	0.029	0.44	1.4

results feed the condition-marginal components $u_{\text{repeat}}(f)$, $SD_{\text{unit}}(f)$, and $SD_{\text{length}}(f)$ in Table I. This is to quantify a real-world, condition-marginal spread by aggregating within-condition repeatability, across-unit variability, and across-length variability in addition to per-condition repeatability:

$$u_c^2(f) = u_{\text{repeat}}^2(f) + SD_{\text{unit}}^2(f) + SD_{\text{length}}^2(f), \quad (5)$$

$$U_{95}(f) = k_{0.95}(f)\, u_c(f). \quad (6)$$

At each frequency f, $u_{\text{repeat}}(f)$ is the Root Mean Square (RMS) of the per-condition Type A standard uncertainties $u_A(f)$ across Speakers A–D and the three chosen loads. $SD_{\text{unit}}(f)$ is the SD across speakers of condition-averaged means, and $SD_{\text{length}}(f)$ is the RMS across speakers of the SD across the three chosen acoustic load conditions.

The resulting combined standard uncertainty $u_c(f)$ and expanded uncertainty $U_{95}(f)$ of $|Z|$ are summarized in Table I. The expanded uncertainty corresponds to an approximate 95% coverage probability, where $k_{0.95}(f)$ is obtained from a Student's t-distribution using the effective degrees of freedom $\nu_{\text{eff}}(f)$ computed via the Welch–Satterthwaite formula. Here, $\nu_{\text{eff}}(f)$ denotes the effective degrees of freedom associated with the combined standard uncertainty $u_c(f)$. Here, the effective degrees of freedom are primarily governed by inter-unit and load-dependent variability, which are estimated from a fixed number of speakers and load conditions.

The absolute $U_{95}(f)$ values ($\approx 10-11\ \Omega$) mainly arise from inter-unit variability among speakers (SD_{unit}), each having a different voice-coil resistance ($R_e \approx 25-35\ \Omega$), which dominates the combined uncertainty and governs the coverage factor through its finite degrees of freedom. This unit-to-unit offset in R_e causes small shifts in baseline impedance $|Z(f)|$ despite identical acoustic loads. Nevertheless, the CNN model achieved **over 96%** accuracy in acoustic-load prediction, as shown in Section III-D, demonstrating that the learned features $\Delta|Z(f)|$ are robust and independent of speaker-specific parameters. Other Type A uncertainty analyses for the impedance phase, R, and X, and the corresponding condition-marginal results are presented in Tables II–IV.

The Type A uncertainty analysis revealed excellent repeatability in the mean estimates (e.g., u_{repeat} on the order of $10^{-2}\ \Omega$ for $|Z|$ at the selected anchor frequencies) and stable phase response ($< 2.25°$), with inter-unit variability identified as the dominant contributor. Across all parameters ($|Z|$, phase, R, and X), the expanded uncertainties ($U_{95} = k_{0.95}u_c$) were consistent with expected values for electroacoustic impedance measurements using low-impedance transducers, confirming the metrological robustness of the experimental system.

2) Scope and separation of Type B: Type B contributors arise from the hardware specifications of the AD2 measurement system, the series resistor used to infer the current through the DUT, and the measured physical distances. The Type B uncertainties were evaluated in accordance with the GUM by converting published manufacturer specifications into standard uncertainties through explicit assumptions regarding probability distributions and associated divisors. Unless explicitly stated otherwise by the manufacturer, published specifications were interpreted as maximum permissible errors and modeled using rectangular (uniform) probability distributions, with standard uncertainties obtained by division by $\sqrt{3}$. Quantization effects were modeled as uniform over ± 0.5 LSB and converted to standard uncertainties using a divisor of $\sqrt{12}$.

The Type B uncertainty budget is evaluated for the complex impedance $Z(f)$ defined in Eq. 1, which expresses the measurand as a function of the measured voltages $V_1(f)$ and $V_2(f)$ and the series reference resistor R_s. The corresponding effect on the impedance magnitude ($|Z(f)|$) was obtained by applying the same input perturbations to $Z(f)$ and recomputing the magnitude, rather than by treating $|Z(f)|$ as an independent measurand. This approach preserves the physical structure of the measurement model and avoids approximation errors associated with magnitude-only linearization.

Propagation of Type B uncertainties was performed using first-order GUM linearization based on the impedance measurement model in Eq. (1). For this model, the relevant sensitivity coefficients are derived analytically as

$$\frac{\partial Z}{\partial V_1} = \frac{R_s}{V_2}, \qquad \frac{\partial Z}{\partial V_2} = -\frac{R_s V_1}{V_2^2}, \qquad \frac{\partial Z}{\partial R_s} = \frac{V_1 - V_2}{V_2}.$$

Table V summarizes the dominant Type B uncertainty contributions, including the relevant datasheet specifications, assumed probability distributions, associated standard uncertainties, and the sensitivity coefficients used for uncertainty propagation. Each Type B input quantity is denoted by $x_i \in \{V_1, V_2, R_s, L\}$, where u_{x_i} represents the corresponding standard uncertainty obtained from the published specifications using explicitly stated distributional assumptions and appropriate GUM divisors.

Because both V_1 and V_2 are obtained directly from the measured oscilloscope waveforms, the waveform-generator setpoint accuracy does not appear as an independent input quantity in the measurement model and is therefore excluded to avoid double counting. The Type B input quantities V_1, V_2, and R_s are assumed to be mutually uncorrelated for first-order uncertainty propagation. Potential partial correlation between the oscilloscope channels used to measure V_1 and V_2, arising from shared gain or offset calibration, is not explicitly modeled; neglecting these covariance terms is expected to yield a conservative estimate of the combined uncertainty. For completeness, the full-scale input range $V_{\text{FS},k}$ reported in Table V denotes the selected vertical scale of oscilloscope channel k, with all measurements conducted at scales ≤ 0.5 V/div to ensure operation within the specified accuracy limits.

Also, the physical distance L defining the acoustic load (Fig. 1) constitutes a further Type B input quantity. The L was

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set mechanically using the syringe plunger position and verified using a vernier caliper; the associated standard uncertainty u_L was derived from the stated caliper accuracy, assuming a rectangular probability distribution. Although L does not appear explicitly in the electrical impedance expression of Eq. 1, variations in L modify the acoustic boundary condition and thereby influence the measured impedance $Z(f)$. Because the functional dependence $Z(f, L)$ is not expressed in closed analytical form, the corresponding sensitivity coefficient $\partial Z/\partial L$ was estimated empirically from the measured impedance data by finite differencing between adjacent enclosure-length conditions in the vicinity of the operating point.

C. Length Prediction Results Based on Speaker Specifications

In this experimental setup, the model was trained to classify labels incorporating both speaker specification and length identifiers. As presented in Table VI, the model achieved

- Accuracy = 98.21% and F1-Score = 0.9821,

demonstrating strong overall performance. The corresponding confusion matrix for the test set is provided in the online supplementary material, Figure S1(a).

D. Length Prediction Results Independent of Speaker Specifications

In this scenario, the model was assumed to perform classification based solely on length information, requiring it to implicitly learn a unified representation across different speakers. As shown in Table VI, the model achieved

- Accuracy = 96.61% and F1-Score = 0.9662,

indicating a slight performance degradation while still maintaining high effectiveness. The corresponding confusion matrix is provided in the online supplementary Figure S1(b).

E. Length Prediction Results of Individual Speakers

Because speaker-specific characteristics may affect model performance when training on combined data, the model was also trained separately for each speaker to assess whether individualized training improves performance. As shown in Table VII, the model demonstrated high performance across all settings, achieving 100% accuracy for speakers B, C and D. Although speaker A exhibited the lowest performance, with

- Accuracy = 92.86% and F1-Score = 0.9305,

it still remained highly effective, indicating the reliability of both the collected dataset and the proposed model. The confusion matrices of the individual speakers are presented in the online supplementary Figure S2.

IV. Discussion

A. Explaining the ML Model Through Motional Impedance

The downward shift of the electrical impedance peak frequency is observed when an additional acoustic volume is attached to an electroacoustic transducer. It arises from mechanical–acoustic coupling, in which the added acoustic load

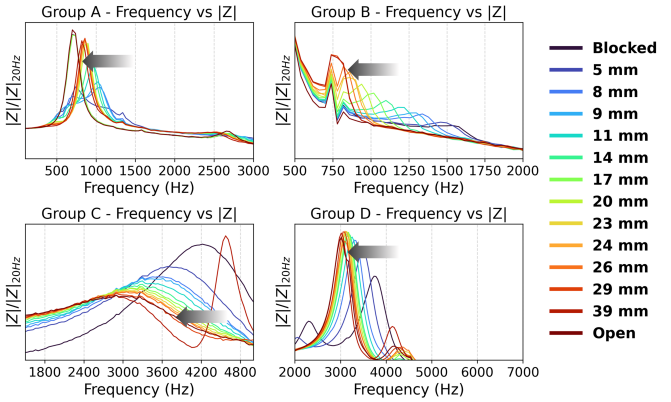


Fig. 6: Zoomed-in view of Figure 3, illustrating the shifts in electrical impedance resonance frequency (gradient arrow) with increasing tube lengths (from dark blue to dark red).

alters the interaction between the diaphragm mass (M) and the total system compliance (C_{total}), thereby shifting the resonance frequency (f_r) as in $f_r = \frac{1}{2\pi\sqrt{MC_{\text{total}}}}$. When the acoustic volume (V) increases, the acoustic compliance (C_a) also increases, as compliance is directly proportional to volume:

$$C_a = \frac{V}{\rho c^2}, \tag{7}$$

where ρ is the air density and c is the speed of sound [22]. This increase in compliance reduces the overall stiffness of the system, increasing the C_{total} and lowering the resonance frequency. At this lower resonance frequency, the transducer diaphragm reaches its maximum vibrational amplitude, which in turn maximizes the *motional impedance* [8].

Figure 6 provides experimental evidence supporting this behaviour. The gradient arrow indicates the direction of the frequency shift as additional tube lengths are introduced, while the darkening gradation highlights the magnitude of these changes. This downward shifting of acoustic impedance signature when increasing tube lengths is also observed in Fig. 3 and Fig. 5. We observed this trend, which supports the length prediction results and reveals both general and speaker-specific behaviors. The overall frequency shift pattern is similar or remains consistent across different speakers (see III-D), while depending on the speaker type, the model varies in the magnitude and rate of these shifts (see III-C).

The electrical impedance peak typically occurs near the mechanical resonance frequency, where the system's reactance components (mass reactance $X_m = \omega M$ and compliance reactance $X_c = 1/\omega C$) cancel out. At that frequency point, X_c and X_m, only the resistivity dominates the total impedance, such as electrical and mechanical [23], [24]. In other types of electroacoustic, piezoelectric ceramics, the impedance spectrum exhibits a distinct peak at the resonance frequency. It corresponds to the point of maximum efficiency in converting electrical energy into mechanical motion [25]. For loudspeakers, the impedance curve exhibits a peak at resonance frequency, indicating the point at which the diaphragm vibrates most efficiently and the system transitions between stiffness-controlled and mass-controlled behavior [26]. As acoustic

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TABLE V: Type B uncertainty budget for impedance measurements using the AD2, a resistor R_s (nominally 100 Ω), and the measured length L (Fig. 1), evaluated following GUM recommendations [21]. All sources assume rectangular distributions.

Source	Published specification	Standard uncertainty u_i	Sensitivity coefficient
Oscilloscope channel amplitude accuracy	Accuracy (scale $\leq$ 0.5 V/div, $V_{\text{in,CM}} = 0$): $\pm 10\,\text{mV}$ $\pm 0.5\%$	$u_{V_1} = \frac{10\,\text{mV} + 0.005\,V_{\text{FS},1}}{\sqrt{3}}$, $u_{V_2} = \frac{10\,\text{mV} + 0.005\,V_{\text{FS},2}}{\sqrt{3}}$	$\frac{\partial Z}{\partial V_1} = \frac{R_s}{V_2}$, $\frac{\partial Z}{\partial V_2} = -\frac{R_s V_1}{V_2^2}$
ADC quantization (oscilloscope)	Effective LSB for selected input range	$u_{q,1} = \frac{\text{LSB}_1}{\sqrt{12}}$, $u_{q,2} = \frac{\text{LSB}_2}{\sqrt{12}}$	$\frac{\partial Z}{\partial V_1}$, $\frac{\partial Z}{\partial V_2}$
Series resistor R_s tolerance	Nominal $R_s = 100\,\Omega$, tolerance $\pm 0.1\%$	$u_{R_s} = \frac{0.1\% \times R_s}{\sqrt{3}}$	$\frac{\partial Z}{\partial R_s} = \frac{V_1 - V_2}{V_2}$
Series resistor thermal drift (TCR)	TCR $= 50\,\text{ppm}/^\circ\text{C}$; $\Delta T = \pm 2^\circ\text{C}$	$u_{R_s,T} = \frac{R_s \cdot \text{TCR} \cdot \Delta T}{\sqrt{3}}$	$\frac{\partial Z}{\partial R_s}$
Physical distance L (acoustic load setting)	Vernier caliper accuracy $\pm 0.025\,\text{mm}$ (Fig. 1)	$u_L = \frac{0.025\,\text{mm}}{\sqrt{3}}$	$\frac{\partial Z}{\partial L}$ (empirical)
Combined Type B uncertainty propagation: $u_B^2(Z) = \sum_i \left(\frac{\partial Z}{\partial x_i} u_{x_i}\right)^2$, $U_{B,95} = k_{0.95}\, u_B$			

TABLE VI: Metric Comparison Based on Identifiers

Evaluation Metric	Speaker and Length	Only Length
Accuracy	98.21%	96.61%
F1-Score	0.9821	0.9662

TABLE VII: Metric Comparison on Individual Speaker Data

Evaluation Metric	A	B	C	D
Accuracy	92.86%	100%	100%	100%
F1-Score	0.9305	1.0000	1.0000	1.0000

compliance increases (i.e, when larger acoustic volumes are introduced), the mechanical resonance frequency shifts to a lower value, and the electrical impedance peak follows accordingly.

These patterns emerge from coupled electrical, mechanical, and acoustic interactions. Although traditional physical or signal-processing models can partially capture such behavior, they often degrade in dynamic or noisy environments due to limited flexibility [27], and typically require retuning when physical properties change, limiting scalability.

Another interesting observation is obtained by comparing the acoustic characteristic impedance tube model (Fig. 5) with the measured electrical impedance responses (Fig. 3). Further interpretation of the CNN accuracy results for each speaker (Table VII) reveals that alignment of electrical and acoustic impedance features at frequencies where the acoustic impedance is low and the electrical impedance is high, leads to improved classification accuracy.

CNNs are well-suited for this task due to their ability to capture localized spectral patterns and nonlinear relationships in frequency-dependent impedance data. By stacking nonlinear modules (e.g., ReLU), CNNs can model complex dependencies in the input features [28], [29]. In addition, their overlapping receptive fields enable the extraction of localized structures, which is particularly effective for identifying subtle frequency-dependent variations [30]. These properties support robust classification performance across varying acoustic conditions [31].

B. Measured Features for Intelligent Sensing

While many data-driven sensing systems rely on deep networks to learn feature representations from large datasets, this study leverages the inherent structure of the raw data by using impedance measurements directly as information-bearing features with minimal preprocessing.

This approach exploits the inherent structure and physical meaning already embedded in the measured data, avoiding extensive signal transformation or feature engineering pipelines. Measured features are often domain-specific and interpretable, enabling clearer insight into how sensing decisions are formed. In electroacoustic systems, such features may include impedance magnitude and phase characteristics that directly relate to acoustic loading, compliance, and resonant behavior. Embedding this physical knowledge at the sensing stage simplifies the learning task, supports more compact and efficient model architectures, and reduces computational cost and training time, an important consideration for intelligent sensing at the edge. Measured features are particularly advantageous in scenarios where training data are limited. Unlike purely data-driven representations that typically require large datasets to learn robust abstractions, physics-informed measured features tend to be concise and information-rich. Models trained on such features can achieve stable and reliable performance even with relatively small datasets. Because these features are causally linked to physical system behavior and often pre-filtered and/or preprocessed by the sensing mechanism itself, they can provide cleaner and more stable inputs to the learning model.

C. Discussion of Uncertainty Analysis

The dominant contribution to the absolute uncertainty, however, was the unit-to-unit variation of the transducers themselves, a property ties to manufacturing options such as voice coil types. The range of nominal DC impedance for voice coils in headphones and earbuds is wide from 8 Ω to 600 Ω. Even if we consider parts normally used for portable or mobile applications, the range is usually between 8 Ω to 50 Ω. Because the nominal electrical impedances of the tested

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transducers under acoustic loading were $|Z| \approx 25{-}35\ \Omega$, an absolute expanded uncertainty of $U_{95} \approx 11\ \Omega$ (Type A, inter-unit dominated) is naturally expected and physically reasonable.

The electrical impedance observed at the terminals can be written, following Beranek's electro–mechano–acoustic formulation [32, Eq. (3.62)], as

$$Z_e(\omega) = R_e + j\omega L_e + \frac{(B\ell)^2}{Z_m(\omega)}, \tag{8}$$

where R_e and L_e are the voice-coil resistance and inductance, and $(B\ell)^2/Z_m$ is the mechanical/acoustic branch reflected into the electrical domain. The mechanical impedance $Z_m(\omega)$, in turn, can be expressed in the form given by Olson for an electrodynamic radiator [33, Eq. (6.14)] as

$$Z_m(\omega) = R_{ms} + j\omega M_{ms} + \frac{1}{j\omega C_{ms}} + S_d^2 Z_a(\omega), \tag{9}$$

where R_{ms}, M_{ms}, and C_{ms} are the mechanical loss, moving mass, and compliance, respectively, and the term $S_d^2 Z_a$ makes explicit the transformation of the acoustic load Z_a at the diaphragm surface into the mechanical domain. In many loudspeaker treatments, this load term is absorbed into R_{ms}; here it is written separately to clarify why different acoustic terminations (blocked vs. open) produce different apparent electrical impedances and, consequently, different absolute uncertainties.

A small change in $Z_a(\omega)$ or in the transducer's intrinsic parameters (R_e, $B\ell$) is carried through Eq. 8–9 and appears directly at the terminals. In the present study, uncertainty in the physical distance L defining the acoustic load contributes to uncertainty in $Z_a(\omega)$ and is therefore treated as a Type B input quantity through an empirically estimated sensitivity coefficient $\partial Z/\partial L$. The Type B contribution resulted in a relative combined standard uncertainty u_B on the order of a few percent. The corresponding expanded uncertainty $U_{B,95}$ was obtained using the coverage factor $k_{0.95}$ defined in the uncertainty propagation model.

Together, the Type A and Type B analyses confirm stable, traceable, and physically interpretable measurements linking acoustic loading to electrical impedance.

V. Conclusions

This paper investigates the potential of speakers (specifically earphones), traditionally used as audio output transducers, to function as sensing elements by exploiting their electroacoustic coupling, known as motional impedance. Electrical impedance measurements were acquired under systematically varied acoustic load conditions, representing different insertion depths and port configurations. The results demonstrate that acoustic conditions alter the electrical impedance response, confirming that earphones can serve as self-sensing transducers. Even tiny changes in impedance response due to acoustic changes can be captured owing to data pre-processing and the AI model architecture of our choice, the CNN. A comprehensive metrological evaluation was conducted following the GUM method, quantifying both statistical and systematic uncertainty components to establish measurement traceability and reproducibility.

The current study is limited in its ability to provide continuous parameter estimation. As part of future work, we will further investigate regression-based models to enable continuous cavity length estimation and improve generalization beyond the discrete measurement points.

The present study advances the state of the art in the use of earphones as sensing transducers combined with AI techniques by providing a comprehensive uncertainty analysis of electroacoustic sensing data within an AI framework. To the best of our knowledge, such a systematic integration of metrological evaluation with data-driven modeling has not been previously reported. This study establishes a rigorous foundation for electroacoustic sensing and highlights a promising direction that bridges sensor systems and artificial intelligence, demonstrating how rigorous metrological evaluation can complement data-driven analysis in general.

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